

WORKPACK 4

Advanced Transforms and Add Columns

Level 3 Data Technician · GreenWheels Urban Mobility · Self-Diagnostic Pack

Get Ready to Smash This

A head-start check — this is not a test

Below are eight things this pack covers. Tick the ones you can already do.

If you can tick all eight: jump to the Mini-Mission on page 8 and prove it. Then move on to Pack 5.

- ☐ I can add a Custom Column in Power Query using a formula I write in M language.
- ☐ I know the M formula to divide one column by another and round the result.
- ☐ I can add a Conditional Column that assigns a label based on a value in another column.
- ☐ I can set up more than one condition in a Conditional Column (for example: Low, Medium, High).
- ☐ I can use Group By to produce a summary table showing totals for each zone.
- ☐ I know the difference between Group By (summarise) and Add Column (keep all rows).
- ☐ I can add an Index Column to give each row a unique row number.
- ☐ I can use Replace Errors to handle divide-by-zero or other error values in a column.

Brian in the afternoon session says: "I added a column in the spreadsheet by typing a formula in column K. Is that the same as adding a column in Power Query?" Let's look at why they are different.

What This Pack Is About

Pack 3 showed you how to clean data in Power Query. This pack shows you how to add new columns that calculate, classify, and summarise your data — all inside Power Query, before the data reaches the spreadsheet. These new columns refresh automatically when the data is updated.

You will use Pack4_GW_data.xlsx, which is the clean 40-record GreenWheels dataset. Every exercise uses real values from the data so you can check your answers against the numbers.

Why Add Columns in Power Query Rather Than Excel?

When you add a column in a spreadsheet by typing a formula in a new column, that formula is not part of the Power Query pipeline. If someone refreshes the data connection, the spreadsheet table may overwrite your formula column or shift it out of place. Adding the column inside Power Query means it is always part of the clean, refreshed output.

NOTE

Custom Column — write your own calculation in M language. The column is calculated row by row.

Conditional Column — assign a label or value based on a condition. No code needed — just fill in a form.

Group By — collapse many rows into one summary row per group (for example, one row per zone showing the total hires).

Index Column — add a sequential row number. Useful for sorting, ranking, and lookups.

The Skills

Load Pack4_GW_data.xlsx into Power Query before you start. You will work on the same query for Chunks 1 to 4, then create a separate Group By query in Chunk 5.

1

Custom Column

Writing a calculation in M language

SCENARIO

"I need to know the revenue earned per hire for each record," says Priya.

"That is just Total Revenue divided by Total Hires. Can you add it as a new column in Power Query so it appears every time I refresh?"

A Custom Column is a new column whose values are calculated by a formula you write. Power Query uses a language called M. M formulas look similar to Excel formulas but use square brackets for column names instead of cell references.

WORD TO KNOW

Custom Column

Add Column tab → Custom Column. Opens a dialog where you type a column name and an M formula. The formula runs for every row. Column names are written inside square brackets: [Total Revenue].

WORD TO KNOW

M language

The formula language that Power Query uses. Column names go in square brackets: [Column Name]. Operators are the same as Excel: + - * /. To round a number: Number.Round([value], decimal_places).

To add Revenue Per Hire: go to Add Column → Custom Column. Name the column 'Revenue Per Hire'. Enter this formula in the dialog box:

```
Number.Round( [Total Revenue] / [Total Hires], 2 )
```

This formula divides the revenue by the number of hires and rounds the result to two decimal places. HC-001 should show approximately 8.50 ($4352 \div 512 = 8.50$).

TRY IT

Load Pack4_GW_data.xlsx into Power Query. Go to Add Column → Custom Column. Name the new column 'Revenue Per Hire'. Enter the formula shown above. Click OK.

Check HC-001: Total Revenue = 4352, Total Hires = 512. What does your formula return? Is it 8.50?

Scroll to HC-016: Total Revenue = 4140, Total Hires = 487. What should Revenue Per Hire be? Calculate it by hand first, then compare with what Power Query shows.

CHECK YOURSELF

What would the formula return if Total Hires was 0 for a record?

Answer: It would return an Error — you cannot divide by zero. Power Query shows 'Error' in that cell. To handle this gracefully, you would wrap the formula in a try/otherwise expression, or use Replace Errors after (covered in Chunk 5).

2

Conditional Column

Classifying records with labels

SCENARIO

"I want a Performance Band column," says Marcus.

"Low if Total Hires is under 200, Medium if it is 200 to 400, High if it is over 400. No formulas — I just need a label in a new column."

A Conditional Column lets you assign a text label or value to each row based on a condition. You set it up using a simple form — no formula writing required. Power Query converts your choices into an M formula in the background.

WORD TO KNOW

Conditional Column

Add Column tab → Conditional Column. A form where you build if/else rules: if [Column] meets a condition, then assign this value; otherwise try the next rule. The last row is the 'else' — what happens when no other rule matches.

The three rules for Performance Band:

Condition	Value if true	Example
Total Hires < 200	Low	HC-017: 198 hires → Low
Total Hires <= 400	Medium	HC-025: 287 hires → Medium
Otherwise (else)	High	HC-003: 689 hires → High

To add this column: go to Add Column → Conditional Column. Name it 'Performance Band'. Add the first rule: Column Name = Total Hires, Operator = is less than, Value = 200, Output = Low. Add a second rule for less than or equal to 400, output Medium. Set the else value to High.

TRY IT

Add the Conditional Column using the three rules above. Name it 'Performance Band'. Click OK.

Check these records: HC-017 (198 hires) should show 'Low'. HC-025 (287 hires) should show 'Medium'. HC-003 (689 hires) should show 'High'. Do they?

How many rows are classified as High? How many as Low? (You can check by scrolling, or by looking at the filter dropdown on the new column.)

CHECK YOURSELF

If a record has exactly 200 hires, will it be classified as Low or Medium?

Answer: Medium. The first rule says 'is less than 200' — so exactly 200 does not trigger it. The second rule says 'is less than or equal to 400' — exactly 200 matches this rule, so the record gets 'Medium'.

3**Group By***Summarising data to one row per group***SCENARIO***"I need the total hires per zone for the whole year," says Priya.**"Not 40 individual records — just five rows, one per zone, showing the zone total." That is Group By.*

Group By collapses all rows with the same value in a column into one summary row. You choose what to group on (for example Zone) and what to calculate for each group (for example Sum of Total Hires). The result is a smaller, summarised table.

WORD TO KNOW**Group By**

Home tab → Group By (or Transform tab → Group By). Choose a column to group by, then add one or more aggregation columns: Sum, Average, Min, Max, Count, etc.

WORD TO KNOW**Aggregation**

A calculation that combines many values into one — Sum, Average, Min, Max, Count. Group By uses aggregation to produce one value per group.

To group by Zone: go to Home tab → Group By. Set the group column to Zone. Add a new column called 'Total Hires (Zone)', operation = Sum, column = Total Hires. Add another called 'Avg Satisfaction', operation = Average, column = Customer Satisfaction Score. Click OK.

NOTE

Important: Group By creates a new, summarised query. If you apply it to your main query, you will lose the detail rows. Either: (a) duplicate your query first (right-click the query name → Duplicate), then group the duplicate; or (b) use 'Add Column → Group By' to keep both.

TRY IT

In the Queries pane on the left, right-click your query and choose Duplicate. Rename the duplicate 'Zone Summary'.

With the Zone Summary query selected, go to Home → Group By. Group by Zone. Add: 'Total Hires (Zone)' = Sum of Total Hires, 'Total Revenue (Zone)' = Sum of Total Revenue, 'Avg Satisfaction' = Average of Customer Satisfaction Score.

After clicking OK, how many rows are in the Zone Summary table? What zone has the highest total hires?

CHECK YOURSELF

Why do we duplicate the query before applying Group By rather than applying it directly to the main query?

Answer: Applying Group By to the main query replaces it with the summary table. You lose the 40 individual rows. Duplicating first means you keep the detail query and have a separate summary query — two outputs from the same cleaned source data.

4**Index Column***Giving every row a unique number***SCENARIO***"I want to rank stations from highest to lowest total hires," says Marcus.**"If I sort the data, I want a column that shows position 1 for the top station, position 2 for the next, and so on."*

An Index Column adds a sequential number to each row. Starting from 1 (or 0, or any number you choose), it numbers every row in the order they appear in the query at the moment you add the index.

WORD TO KNOW**Index Column**

Add Column tab → Index Column → From 1 (or From 0, or Custom). Adds a whole number column that counts from the starting value. The numbers reflect the order of rows at that point in the Applied Steps.

Sorting first, then adding an index, is how you create a rank column. Sort your data by Total Hires descending, then add an Index Column starting from 1. Row 1 will be the station with the most hires; the index number becomes that station's rank.

TRY IT

On your main query (not Zone Summary), sort the Total Hires column descending (right-click the column → Sort Descending).

Go to Add Column → Index Column → From 1. A column called 'Index' appears. Rename it 'Hire Rank'.

Which station is ranked 1? Which is ranked 40? What is the Hire Rank of HC-017 (University Gate Q1 2024)?

CHECK YOURSELF

If you add an Index Column before sorting rather than after, what does the rank column represent?

Answer: It represents the original row order in the file, not the sort order. The index numbers tell you nothing meaningful about performance — they are just row positions 1, 2, 3... in the order the data arrived. Always sort first, then index.

5**Replace Errors***Handling values that break calculations***SCENARIO**

"Our data dictionary says Active Bikes could theoretically be 0 in future data," says Priya.

"If someone runs your Revenue Per Hire formula on a record with 0 hires, it will error. Can you make it show 0.00 instead of Error?"

When a calculation produces an error — divide by zero, wrong data type, missing value — Power Query shows the word 'Error' in that cell. Replace Errors lets you swap every error in a column with a fixed value of your choice.

WORD TO KNOW**Replace Errors**

Right-click a column header → Replace Errors. Enter the value to use in place of any errors in that column. Common replacements: 0 for numeric errors, null to mark the cell as blank, or a text note like 'Check data'.

WORD TO KNOW**try ... otherwise**

The M formula pattern for handling errors in a Custom Column. Example: try [Total Revenue] / [Total Hires] otherwise 0. If the division succeeds, use the result. If it errors, use 0 instead.

There are two ways to handle errors. Option A (quick): apply Replace Errors after adding the column. Option B (robust): write the try/otherwise pattern inside the Custom Column formula itself.

Option B — inside the Custom Column formula:

```
try Number.Round( [Total Revenue] / [Total Hires], 2 ) otherwise 0
```

TRY IT

Go back to the Revenue Per Hire step in Applied Steps. Right-click it and choose 'Edit Settings' (or double-click the step). Add 'try ... otherwise 0' around the existing formula as shown above.

To test it: temporarily set Total Hires for HC-001 to 0 in the source file, then Refresh the query. Does Revenue Per Hire show 0.00 for that row instead of Error? (Change the value back afterwards.)

CHECK YOURSELF

What is the difference between Replace Errors and try/otherwise in a Custom Column formula?

Answer: Replace Errors is applied after the column is created — it replaces any errors that already exist. try/otherwise is built into the formula itself — it prevents the error from occurring in the first place and handles it gracefully for every future refresh.

Spot the Mistake

Can you see what went wrong?

Brian added a Conditional Column for Performance Band. His rules: Total Hires less than 400 = Medium, Total Hires less than 200 = Low, otherwise = High.

When he checks the results, every record shows either Medium or High. No row shows Low — even HC-017 with only 198 hires.

WATCH OUT

The rules are in the wrong order. Power Query checks each rule from top to bottom and stops at the first match. Total Hires < 400 catches all rows under 400 — including those under 200 — before the Low rule is ever reached. The Low rule never fires. Rules must go from most specific (smallest threshold) to least specific.

CHECK YOURSELF

What is the correct order for the three Performance Band rules?

Answer: First: Total Hires less than 200 → Low. Second: Total Hires less than or equal to 400 → Medium. Else: High. This way, records under 200 are caught first. The rest that reach rule two are in the 200–400 range. Everything above 400 falls through to High.

Mini-Mission

Two minutes — prove it

Load Pack4_GW_data.xlsx into Power Query. Without referring back to the chunks, add: (1) a Revenue Per Hire column using try/otherwise 0, (2) a Performance Band conditional column with three correct rules, (3) a Zone Summary duplicate query showing total hires and average satisfaction per zone.

Close and load both queries. Check that your loaded tables have: 40 rows in the main table, 5 rows in the Zone Summary.

You Can Now Answer

These are the kinds of questions that come up in Scenario Demonstration 2

You should now be able to answer all of the following:

- ☐ What is a Custom Column in Power Query, and how do you write an M formula?
- ☐ What does Number.Round do, and why is rounding important when dividing?
- ☐ What is a Conditional Column, and why does rule order matter?
- ☐ What does Group By do, and what is the difference between Group By and Add Column?
- ☐ How do you create a rank column using sort order and an Index Column?

- ☐ What is the difference between Replace Errors and try/otherwise?

If You Got Stuck

No problem — here is what to practise next

The first time I wrote an M formula I put the column name in inverted commas instead of square brackets. It just returned the same text for every row. It takes one or two mistakes before it clicks.

— **Marcus, Level 3 Data Technician cohort 2024**

If Custom Column felt tricky: try writing a simpler formula first — for example, [Total Hires] * 2 — just to see that square brackets refer to columns. Then build up to the division with rounding.

If Conditional Column rule order confused you: draw it as a flowchart before setting it up. Start with the most specific condition at the top. The else at the bottom catches everything that did not match any rule.

Pack 5 covers Merge and Append — combining two separate datasets into one. You will use the Group By query you built here as part of that pack.